KONX: A Dynamic Approach for Explainable AI in Higher Education

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ABSTRACT

In the era of data-driven decision-making, Higher Education Institutions (HEIs) can greatly benefit from the potential of explainable Artificial Intelligence (XAI) to provide transparent and interpretable insights. This paper presents the KONX (CONNECTS) approach, a methodology that leverages semantic web technologies to create a dynamic and comprehensive knowledge graph for advanced predictive models in academic advising. The KONX methodology focuses on harmonizing heterogeneous educational data sources, enabling seamless data querying and manipulation. By incorporating a feedback mechanism, the KONX approach remains adaptable to changes in the academic domain, continuously updating and maintaining its knowledge representation. To practically apply and evaluate the proposed methodology, a prototype was implemented and tested on an experimental case study concerning student outcomes prediction. The implemented prototype includes a graphical SPARQL generator interface to streamline the construction of SPARQL queries in an integrated way. In this way, this paper proposes both a comprehensive XAI methodology and a holistic technological infrastructure for applying the methodology in real-time scenarios. By bridging the gap between AI decision-making and human-comprehensible explanations, the KONX approach enhances transparency and user trust in AI-driven systems in the education sector.

CCS CONCEPTS

• **Computing methodologies**; • **Artificial intelligence**;

KEYWORDS

XAI, Semantic Web Technologies, Academic Advising, Higher Education

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1 INTRODUCTION

In the rapidly evolving field of AI, the integration of predictive modeling techniques in various applications has become increasingly prevalent. In the education sector, AI can assist learners, educators, and Higher Education Institutions (HEIs) managers in understanding student parameters, learning trajectories, and data insights in the age of "big data" [1]. Modern AI approaches can be utilized to provide consistent predictive analysis and decision support for stakeholders, processing specific student parameters derived from the learner model to produce useful recommendations [2]. However, the lack of interpretability in many AI approaches creates a disconnect between researchers and end-users [3]. An AI-enhanced system is often referred to as a black box because end-users struggle to easily interpret the calculations employed by the algorithms [4] [5].

To address these issues, this paper presents the KONX methodology, aiming to enhance user understandability in AI-based Academic Advising Systems (AAS) and foster increased trust and transparency. The KONX approach utilizes a knowledge graph represented as an ontology to describe domain knowledge in the academic advising context. Aligned with the KONX ontological schema, data derived from multiple heterogeneous educational sources are consolidated and harmonized within a triplestore database, enabling deductive reasoning and querying. By leveraging RDF triples, the KONX methodology unveils the rationale behind prediction systems, making the explanations more interpretable and insightful. Moreover, the feedback mechanism in the KONX approach enables continuous learning and improvement, allowing the system to adapt and evolve based on changing domain requirements. This solution will be seamlessly integrated into the existing AAS EDUC8EU, which is powered-up by AI, thereby enhancing its capabilities to generate more transparent, and easily interpretable recommendations for students [6].

Table 1: SWOT Analysis

Strengths	Weaknesses
• XAI can help educators and administrators understand the factors that influence student success and make data-driven decisions to improve student outcomes. • XAI can provide interpretable models that are easy for educators to understand and use. • It can help to build trust and understanding among stakeholders, by providing a clear and detailed explanation of the model's decision-making process.	• XAI technologies are still relatively new, and there may be a lack of expertise and resources to implement them in educational settings. • Developing XAI models can be more complex and computationally expensive than traditional AI models. • It can be difficult to determine the "right" level of interpretability for a given application, which may lead to trade-offs between interpretability and performance.
Opportunities	**Threats**
• XAI can be used to create new educational products and services that provide targeted support to students. • XAI can be used to help solve complex problems that are difficult to solve with traditional AI methods, such as detecting bias or identifying causal relationships.	• There may be resistance from administrators (and educators) to using XAI if they are not fully understood and trusted. • The field of XAI is rapidly evolving, and the technology may change quickly, making it difficult to stay up-to-date with the latest developments.

The paper proceeds with an overview of the related work in Section 2, where existing research on XAI is explored. In Section 3, we delve into the details of the KONX approach, presenting the methodology's key components and semantic model development process. The technical architecture and implementation details of the KONX prototype are elaborated in Section 4, showcasing its robust and versatile design. Following this, Section 5 introduces the implemented SPARQL Designer Tool, which streamlines the creation and management of SPARQL queries during the development phase. In Section 6, we present the experimental case study that practically applies and evaluates the KONX methodology. The conclusions and future work are discussed in Section 7, highlighting the contributions made in this research and providing insights into potential enhancements and evaluations for the future.

2 MOTIVATION AND RELATED WORK

2.1 Motivation

The field of predictive analysis in education draws from an array of educational data sources, creating a landscape of diverse and heterogeneous information. These sources encompass a wide range of data, including records from traditional courses within HEIs, academic advising sessions, and data generated through e-learning systems, especially with the accelerated adoption of online learning during events like the COVID-19 pandemic [7]. One of the key challenges in leveraging these diverse data sources is the presence of semantic inconsistencies among these sources [8]. Different sources may use distinct terms, formats, and types to represent similar concepts, leading to potential difficulties in integrating and interpreting the data for predictive analysis.

To overcome these challenges and unlock the full potential of the rich educational data, a common semantic model is essential. By establishing a shared understanding of concepts and their relationships, this semantic model will facilitate the fusion of heterogeneous data, ensuring consistency and coherence in the analysis process. In this way, the creation of a common semantic model will enable the harmonization of data from various sources, enhancing data interoperability and allowing for seamless integration of educational data for predictive analysis.

2.2 Related Work

XAI possesses several significant strengths when applied to educational settings. Firstly, it empowers stakeholders by providing them with a better understanding of data-driven insights that influence student success, thereby enabling them to make informed decisions [3]. XAI offers interpretable models that make AI algorithms easier for educators to comprehend and utilize, fostering a more transparent decision-making process. Moreover, by providing clear explanations for the output, XAI helps build trust and understanding among all stakeholders, promoting greater acceptance and collaboration [9].

XAI's application in educational products and services could revolutionize how targeted support is provided to end-users (students and/or HEI managers), addressing their specific needs and enhancing the overall experience [10]. Furthermore, XAI's potential to tackle complex issues, such as detecting bias and identifying causal relationships, offers a promising avenue for addressing longstanding educational challenges [11].

Despite its promise, XAI faces certain challenges within the educational domain. One key weakness lies in the novelty of XAI technologies, which may hinder implementation due to a lack of expertise and resources [12]. Additionally, developing XAI models can sometimes prove to be more complex and computationally expensive compared to conventional AI models, potentially imposing resource constraints [13]. Striking the right balance between interpretability and performance can be challenging, as overly interpretable models may sacrifice predictive accuracy, and vice versa.

Several threats must be carefully considered when integrating XAI into educational contexts. Primarily, the resistance of stakeholders to embrace XAI could hinder its widespread adoption if the technology is too new [3]. Addressing this challenge requires

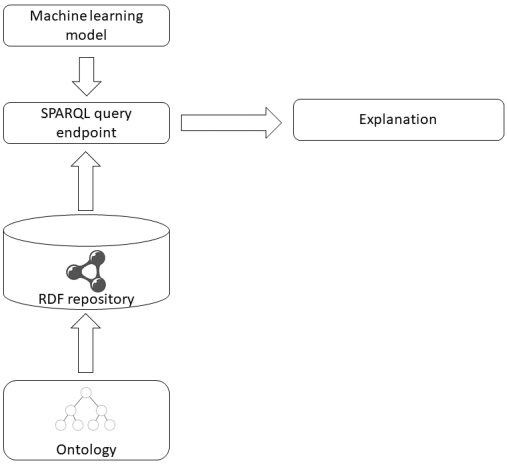

Figure 1: General overview of the proposed methodology

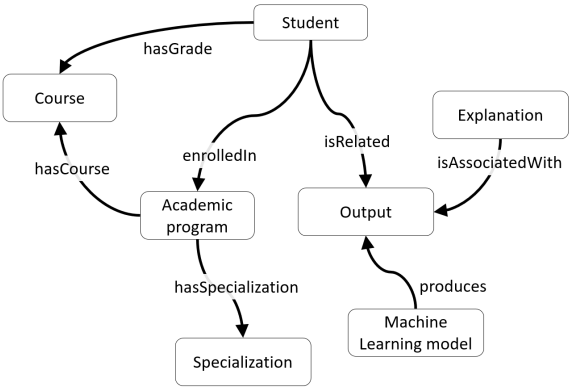

Figure 2: KONX ontology abstract diagram

effective communication and awareness building. Lastly, the rapidly evolving nature of the XAI field may lead to the swift obsolescence of existing technologies and methodologies, making it essential for educational institutions to remain abreast of the latest developments to maximize XAI's potential [14].

Table 1 presents a SWOT analysis of applying XAI to enhance predictive analytics in academic advising.

3 KONX METHODOLOGY

The KONX methodology is a comprehensive approach designed to enhance predictive analytics in academic advising by leveraging XAI. Its core foundation lies in the creation of a knowledge graph, which serves as a complementary tool for representing context and domain-specific knowledge. In the initial stage, the KONX methodology carefully captures and defines the domain-specific knowledge in the form of an ontology. This ontology, tailored to the machine learning domain, acts as a structured framework that establishes meaningful connections between different academic advising concepts, machine learning outputs and their underlying relationships.

Figure 1 illustrates a comprehensive overview of the KONX methodology. Building upon the ontology, the KONX methodology incorporates entities specific to the academic advising domain. These entities are semantically annotated, ensuring compatibility with the ontology's structure, and are stored as an RDF triples in a Triplestore Database. The centralization of knowledge in a common repository offers numerous advantages, such as improved data accessibility, consistency, and ease of maintenance. Furthermore, this approach facilitates the implementation of a SPARQL query endpoint, allowing seamless access to specific pieces of knowledge. The KONX application can utilize this endpoint to trigger specialized queries together with the output of the machine learning models, enabling the retrieval and presentation of targeted explanations to users. This enriches the output with transparent and comprehensible insights.

An essential aspect of this methodology is its incorporation of a feedback mechanism in each execution cycle, generating a feedback message containing new facts. These newly acquired facts play a crucial role in updating the underlying knowledge base. This continuous integration of new knowledge ensures the perpetual maintenance and relevance of the explanations provided by the proposed approach. By embracing a feedback-driven approach, the KONX methodology ensures its adaptability and responsiveness to changes in the academic advising domain.

4 THE KONX SEMANTIC MODEL

The approach used to develop the KONX conceptual model followed the Ontology 101 development process proposed by [20]:

- *Define the domain:* The first step is to clearly define the domain for which the ontology is being developed. The purpose of the KONX model is to represent the knowledge required for XAI in the context of academic advising, capturing concepts related to predictive analytics and their interrelationships. The goal is to enable KONX to provide meaningful academic recommendations based on machine learning analysis.
- *Identify the concepts*: In step 2, all the concepts within the domain were identified. At the upper level, the key concepts and entities include Student, Course, Academic program, Specialization, Output, Explanation and Machine Learning model concepts (Figure 2).
- *Identify the relationships*: All the relationships between the concepts were defined such as Student enrolledIn Academic Program, Student hasGrade in Course etc.
- *Create the Ontology Schema (Classes and Properties)*: In this step, a formal representation of the concepts and relationships was created. Important machine learning concepts, such as feature importance scores for variables in Random Forest Models, coefficient values for most influential attributes in Logistic Regression Models or Local Interpretable Model-agnostic Explanations (LIME) in Neural Network models, were also defined.
- *Populate the ontology*: Once the schema was finalized, we populated the ontology with instances of the concepts and

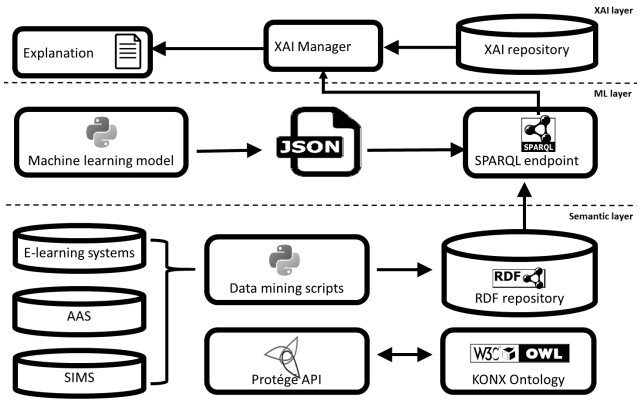

Figure 3: KONX technical architecture

the relationships along with the appropriate data types that were assigned to properties.

- *Verify the consistency*: The consistency of the ontology was verified to ensure that it is free of contradictions and that all relationships are logical and accurate. During this step we ensured that the classes and properties are arranged hierarchically, and applied constraints (e.g., cardinality restrictions) to define the relationships accurately.
- *Test and validate the ontology*: Finally, the ontology was validated by running sample queries against the model and verifying that the expected results are obtained. The necessary refinements were made where required.

5 THE PROTOTYPE

Figure 3 illustrates the technical architecture of the implemented KONX infrastructure. The software architecture is composed of three distinct layers: the semantic layer, the machine learning layer, and the XAI layer. At the heart of the semantic layer lies the comprehensive KONX ontology, which holistically conceptualizes key machine learning features, encompassing the multi-faceted structure of the machine learning domain. To develop the KONX ontology, the Protégé visual interface, in combination with Web Ontology Language 2, was employed at the lower layer.

Within the KONX ontology, data from diverse educational sources can be harmonized and queried comprehensively, ensuring a unified and coherent knowledge infrastructure (Figure 4). The instances, in the form of RDF triples, were diligently stored within a GraphDB repository. To facilitate the process of translating heterogeneous data into RDF triples aligned with the KONX schema, a set of data mining scripts was meticulously developed using Python. This translation process played a pivotal role in ensuring that the data was efficiently integrated into the KONX schema. Once the harmonized data was represented as RDF triples, a SPARQL endpoint can be triggered by the XAI manager component to serve as the gateway for querying the RDF data, regardless of its initial type, source, or syntax. At the same time, the XAI manager component parses the JSON file produced by the machine learning model to complement the SPARQL queries. The result of the specific interoperation is the carefully curated explanations to be presented to

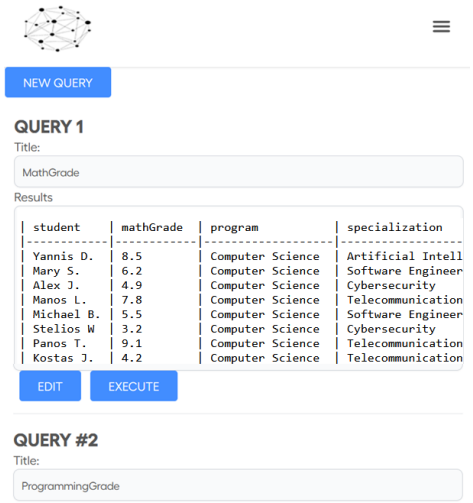

Figure 4: the KONX software environment

the end user, enhancing the interpretability and transparency of the machine learning output.

One important aspect of this approach is its adaptability and extensibility. In any execution, the feedback mechanism has the capacity to extend the KONX knowledge base by generating additional knowledge, requiring only minimal reconfiguration of the semantic model to accommodate new requirements. This flexibility allows the KONX methodology to evolve and incorporate new sources of knowledge and information, ensuring that the system remains up-to-date and capable of handling emerging data streams effectively.

6 AN EXPERIMENTAL CASE STUDY BASED ON THE KONX METHODOLOGY

The methodology proposed in this paper was practically applied and tested in an experimental case study following the work presented by [15]. The specific research work sought to harness the potential of machine learning as an assistive tool to explore educational data repositories, aiming to derive valuable insights for academic advising, improvement, and identification of academically at-risk students. To achieve this goal, a two-phased model was employed, integrating both unsupervised and supervised techniques to capitalize on the benefits of student clustering while addressing prediction-related challenges. In the first phase of the model, the focus was on discovering coherent clusters of students within the data. Subsequently, in the second phase, supervised machine learning models were developed to make predictions for student outcomes, with each cluster being handled individually. For instance, a regression model was trained to predict the Time To Degree (TTD). Figure 5 provides an illustration of the two-phase machine learning model.

The evaluation analysis demonstrated that the two-phase model outperformed traditional approaches in terms of prediction accuracy. However, a significant challenge arose during the practical implementation—while achieving improved predictions, the lack

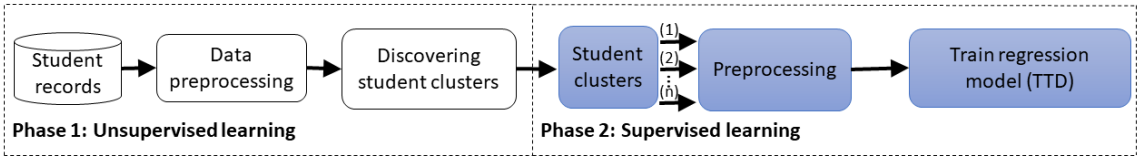

Figure 5: Two-phase machine learning model

of human interpretable results hindered the wider adoption of the approach by HEIs. We conducted a case study, employing the KONX system, utilizing the identical datasets as detailed in paper [15]. In the experimental scenario, the two-phase machine learning model was executed to predict student outcomes. Initially, the K-means algorithm was applied in the first phase to identify possible groups of learners that optimally partitioned the dataset. After executing the algorithm, it was found that three potential clusters of students best separated the dataset. In the second phase, the Random Forest ensemble learning method was employed for regression, producing the mean prediction of individual trees. The Permutation Feature Importance (PFI) technique was utilized to evaluate the importance of features, revealing significant variations in importance across different student clusters. Subsequently, the KONX ontology was employed, providing the essential conceptualization of the machine learning domain. All entities involved in the two-phased machine learning model were instantiated as instances of the ontology, forming the knowledge base for the experimental scenario. The data was then transformed into RDF triples and stored in a repository, facilitating the answering of queries and the production of human-centric explanations in conjunction with the results.

As an example, Query 1 was executed using the KONX system to provide meaningful academic explanations for students belonging to the first cluster, based on the machine learning results. Query 1 first matches all instances of students and their math grades. It then checks the importance of the math grade feature using the MathGrade_PFI received by the machine learning model and categorizes the math grade into different ranges (high, medium, or low) based on predefined thresholds. The query binds the student's program and specialization to relevant variables. Finally, it creates an explanation for each student based on the output acquired by the machine learning model.

With the system's help, stakeholders can leverage this meaningful information to identify students who may benefit from additional academic assistance or intervention. The explanations offered a clear understanding of students' insights and highlighted the meaning of specific machine learning outputs. By utilizing XAI techniques, HEIs can make data-driven decisions to enhance academic advising and support initiatives, ultimately fostering student success within the university.

SPARQL Query 1

```
SELECT ?student ?mathGrade ?program ?specialization
?explanation WHERE {
  ?student :hasGrade ?mathGrade .
  ?student :hasCluster ?Cluster .
  BIND(IF(?MathGrade_PFI > 0.80 && ?Cluster = 1,
        :high_importance,
        :low_importance) AS ?MathGradeImportance)
  BIND(IF(?mathGrade > 7,
        :high_math_grade,
        IF(?mathGrade > 5 && ?mathGrade < 7,
          :medium_math_grade,
          :low_math_grade)) AS ?MathGradeRange)
  ?student :belongsToProgram ?program .
  ?student :enrolledInSpecialization ?specialization .
  BIND(IF(?MathGradeRange = :high_math_grade &&
?MathGradeImportance = :high_importance,
        CONCAT("Student: ", ?student, ", Math Grade: ",
?mathGrade, ". This student's high math grade (> 7) and
importance indicate exceptional mathematical abilities. As part of
the ", ?program, " program, we recommend exploring advanced
math courses to further develop their skills in ", ?specialization, "."),
        IF(?MathGradeRange = :medium_math_grade &&
?MathGradeImportance = :high_importance,
          CONCAT("Student: ", ?student, ", Math Grade: ",
?mathGrade, ". This student's math grade is between 5 and 7,
showing potential in mathematics. As part of the ", ?program, "
program, consider offering additional support to improve their
understanding in ", ?specialization, "."),
          IF(?MathGradeImportance = :high_importance,
            CONCAT("Student: ", ?student, ", Math Grade: ",
?mathGrade, ". This student's math grade is below 5, indicating
some challenges in mathematics. As part of the ", ?program, "
program, we recommend providing focused math tutoring sessions
to enhance their grasp of the enrolled specialization ",
?specialization, ".")))
      AS ?explanation)}
```

7 SPARQL DESIGNER TOOL

The SPARQL designer tool was implemented as an add-on to streamline the creation and management of SPARQL queries during the development phase. With its user-friendly graphical user interface (GUI) and functionalities, the tool empowers users to construct SPARQL queries with fewer inconsistencies, ambiguities, and duplicate patterns. The GUI of the SPARQL designer is depicted in Figure 6 which displays a SPARQL query under creation.

To facilitate query construction, users can use elements originated from the KONX ontology in the design area, thus ensuring consistency in semantic terms. One of the features of the SPARQL designer is its real-time query updates. As users interact with the GUI, the tool instantly reflects their actions, ensuring a seamless and

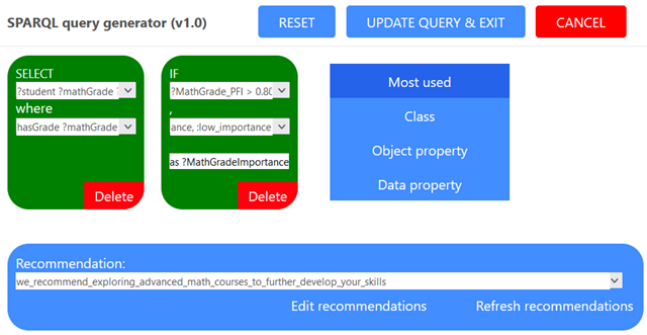

Figure 6: SPARQL query generator

responsive query-building experience. Furthermore, the SPARQL designer includes an intelligent error detection mechanism, detecting and notifying users of lexical and syntactic errors, enhancing the quality of the constructed queries.

8 OUR CONTRIBUTION

In light of the analysis presented, this research work has achieved several key contributions to the field of explainable AI in the education domain:

- *KONX XAI Methodology:* We have introduced a comprehensive methodology for dynamically explaining AI decision-making processes by leveraging a contextual knowledge graph, represented as a semantic model. This approach conceptualizes entities within the AI model and defines the logic for utilizing domain knowledge to analyze the output data of machine learning algorithms. By integrating contextual information into the explanation process, we enable a more transparent and interpretable understanding of AI-driven recommendations in the academic advising domain.
- *Technological Infrastructure for the KONX methodology*: To evaluate our methodology we have designed and implemented a prototype application. The implemented technological infrastructure comprises essential mechanisms, applications, and services that are crucial for supporting the maintenance and management of XAI mechanisms based on the KONX methodology.
- *Comprehensive Case Study Evaluation: We* have conducted an extensive and representative case study to evaluate the effectiveness of our proposed methodology. This real-world evaluation has enabled us to draw valuable insights and identify optimization opportunities for generating explainable AI models in the domain of our interest.

Overall, these contributions mark a significant step forward in promoting interpretability, and user trust in AI-driven systems. The KONX approach, while initially developed within the education domain, holds the potential for seamless extension to various other domains as well. Its underlying foundation, based on a dynamic knowledge graph, provides a versatile framework that can accommodate diverse data sources and contexts.

9 CONCLUSIONS AND FUTURE WORK

In this paper, we introduced the KONX approach, a comprehensive methodology that harnesses the power of XAI in academic advising through the creation of a dynamic knowledge graph. The KONX methodology, built on semantic web technologies, enables the representation of domain-specific knowledge and context, facilitating transparent and interpretable insights into AI-driven output in HEIs. Throughout our research, we demonstrated the effectiveness of the KONX approach in creating meaningful academic recommendations. By incorporating a feedback mechanism, the approach exhibits adaptability to changes in the academic domain, ensuring its knowledge representation remains relevant and up-to-date. The experimental case study conducted to evaluate the methodology showcased the potential of XAI to enhance traditional AI approaches. Moreover, in order to facilitate the maintenance of the SPARQL queries in an integrated way by the domain knowledge experts, a graphical SPARQL generator interface was implemented. This graphical interface streamlines SPARQL query construction and enhances knowledge acquisition for the end-users.

In the future, we aim to further improve the KONX system's effectiveness, by exploring the enrichment of the semantic infrastructure. This involves integrating additional educational data sources and expanding the domain-specific ontology to capture a broader range of machine learning concepts and relationships. Additionally, we plan to investigate the adaptability of the KONX approach in diverse domains, such as healthcare, finance, and e-commerce. By exploring these opportunities, we aim to demonstrate the versatility and robustness of the methodology, further enhancing its potential for real-world applications beyond the education sector.

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